Notebook Run Order

Client Handoff Guide for Final Modeling Notebooks

Purpose

This document explains the recommended order for opening and running the final notebooks included in the client handoff package. The order starts with the core technical deliverables, then moves to the client demo notebooks. Appendix notebooks are optional background materials and do not need to be run for the main handoff results.

Required Dataset

The final notebooks assume the cleaned dataset is named final_validated_dataset.csv. Place this file in the 02_Data folder, or in the same folder as the notebook being run. The final notebooks include fallback path logic to look in both locations.

Recommended Run Order

Order	Notebook	Folder	Purpose
1	TaskA_Current_GSTD_Classification_FINAL.ipynb	03_Final_Notebooks	Core technical notebook for Task A. It evaluates current GSTD classification and compares full, practical, Client-6, LAID, and time-series feature versions.
2	TaskB_7Day_GSTD5_Alert_FINAL.ipynb	03_Final_Notebooks	Core technical notebook for Task B. It builds the 7-day GSTD 5 alert target, evaluates the at-risk Stage 1 framing, compares LightGBM feature sets, checks robustness, and selects LightGBM operating thresholds.
3	Client_Demo_TaskA_Current_GSTD_Classification_FINAL.ipynb	05_Client_Demo_Materials	Client-facing walkthrough for Task A. Use this for presentation or review after the technical Task A notebook is locked.
4	Client_Demo_TaskB_7Day_GSTD5_Alert_FINAL.ipynb	05_Client_Demo_Materials	Client-facing walkthrough for Task B. It includes the final LightGBM demo, threshold interpretation, and example alert behavior.

How to Use the Notebooks

- Run Task A first if the client wants to understand current GSTD classification.
- Run Task B second because it is the main early-warning alert model for GSTD 5 entry within 7 days.
- Use the demo notebooks when presenting the work live. They are shorter, more visual, and easier to explain than the full technical notebooks.
- Use the HTML demo files when the client wants to view the results without running Python or Jupyter.
- Use Dataset_Creation_and_Validation_Workflow.ipynb when the client wants to verify or recreate the final validated CSV from the raw DSSAT outputs.
- Use other appendix notebooks only for background research, exploratory analysis, or explanation of earlier modeling decisions.

Expected Final Outputs

- Task A final model comparison table and recommendation for current GSTD classification.

- Task B final model comparison showing Client-Feasible TS LightGBM as the core recommended model and Client-6 + LAID + TS as an optional extension.
- Task B LightGBM operating threshold summary. In the finalized run, the Client-Feasible TS LightGBM Best F1 threshold is 0.9984.
- Task B alert interpretation showing caught transitions, missed transitions, false alerts, and alert rate.

Appendix Notebooks

The appendix notebooks document the project research trail. Dataset_Creation_and_Validation_Workflow.ipynb is the main appendix file for understanding how the final DSSAT-based dataset was parsed, cleaned, merged, validated, and saved. Other appendix notebooks provide background on early EDA, preliminary LightGBM exploration, and parser utilities.

Final Note

The final notebooks are intended to document the analysis and support client handoff. They are not packaged production software. Before operational deployment, the models should be recalibrated and validated on the client's final real-world or operational dataset.